Relu activation function.

```
import numpy as np
from sklearn.datasets import
load_breast_cancer
from sklearn.model_selection import
train_test_split

data = load_breast_cancer()
X = data.data
Y = data.target
X_train, X_test, Y_train, Y_test = train_test_split(
    X, Y, test_size=0.2, random_state=42)
X_min = X_train.min(axis=0)
X_max = X_train.max(axis=0)
X_train = (X_train - X_min) / (X_max - X_min)
X_test = (X_test - X_min) / (X_max - X_min)
np.random.seed(1)
input_neurons = 30
hidden_neurons = 8
output_neurons = 1
W1 = np.random.rand(input_neurons, hidden_neurons)
b1 = np.zeros(hidden_neurons)
W2 = np.random.rand(hidden_neurons, output_neurons)
b2 = 0.0
Learning_rate = 0.01
epochs = 5
def sigmoid(x):
    return 1 / (1 + np.exp(-x))
def relu(x):
    return np.maximum(0, x)
def relu_derivative(x):
    return (x > 0).astype(float)
print("TRAINING PHASE (Relu Hidden Layer)")
print("=" * 60)
for epoch in range(1, epochs+1):
    Correct = 0
    for i in range(len(X_train)):
        Z1 = np.dot(X_train[i], W1) + b1
```

```
a1 = relu(z1)
z2 = np.dot(a1, W2) + b2
output = sigmoid(z2)[0]
prediction = 1 if output >= 0.5 else 0
error = y_train[i] - output
d_output = error * output * (1-output)
d_hidden = d_output * W2.flatten() * relu_derivative(z1)
W2 += learning_rate * d_output * a1.reshape(-1,1)
b2 += learning_rate * d_output
W1 += learning_rate * np.outer(x_train[i], d_hidden)
b1 += learning_rate * d_hidden
if prediction == y_train[i]:
    correct += 1
accuracy = (correct / len(x_train)) * 100
print(f"Epoch {epoch} | Training Accuracy: {accuracy:.2f}%
correct = 0
for i in range(len(x_train)):
    a1 = relu(np.dot(x_train[i], W1) + b1)
    output = sigmoid(np.dot(a1, W2) + b2)[0]
    prediction = 1 if output >= 0.5 else 0
    if prediction == y_train[i]:
        correct += 1
print("\nFinal Training Accuracy:", (correct / len(x_train)) *
correct = 0
for i in range(len(x_test)):
    a1 = relu(np.dot(x_test[i], W1) + b1)
    output = sigmoid(np.dot(x_test[i], W1) + b1)
    prediction = 1 if output >= 0.5 else 0
    if predictions == y_test[i]:
        correct += 1
print("Final Test Accuracy", (correct / len(x_test)) * 100)
```

OUTPUT

TRAINING PHASE (Relu Hidden Layer)

Epoch 1 Training Accuracy : 62.86%

Epoch 2 Training Accuracy : 62.86%

Epoch 3 Training Accuracy : 62.86%

Epoch 4 Training Accuracy : 62.86%

Epoch 5 Training Accuracy : 62.86%

Final Training Accuracy :
62.85714285714285 4

Final Test Accuracy : 62.28070175438597

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